

User study of omnidirectional treadmill control algorithms in VR

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ABSTRACT

We conducted a comparative user study of the Infinadeck [1] omnidirectional treadmill native control algorithm alongside an approach based on a state observer control scheme combining a kinematic model with error dynamics that allows users to change their walk freely. Based on the results from 22 participants, we observed that the alternative approach outperformed the native algorithm on trajectories involving a left or right turn (with radii of curvature of 0.5m, 1m, and 2m). However, there was no significant difference for straight-line trajectories, and the native approach yielded the best scores for in-place rotations.

1 INTRODUCTION

One key advantage of Virtual Reality (VR) is the ability to experience potentially vast Virtual Environment (VE)s. However, those environments are often much larger than the actual size of the player's current room. This means that the player cannot explore the whole environment using natural locomotion while maintaining a one-to-one mapping between the distance traveled in the real world and the distance traveled in the VR.

Various products leveraging on a sliding platform partially alleviate such limitations by requiring the participant to mimic a walking movement by sliding the feet backward while leaning the body posture forward. However, the resulting experience might be felt as too far from actual walking due to the artificially unbalanced body pose. On the other hand, the Infinadeck offers an omnidirectional treadmill with a floor surface of only 148 x 176 cm. Its control algorithm aims always to bring the user back to the center of the treadmill surface. The main difficulty is to minimize perceptible discontinuities while compensating for user movement. Therefore, we investigated whether an alternative control algorithm could outperform the original algorithm, with in mind to extend it later by anticipating the user dynamics.

2 RELATED WORK

Souman et al. [2] succeeded in compensating for user's displacements during speed variations on a large unidirectional treadmill. This was achieved through a primary feedback loop based on the user's relative position to a reference position and a feed-forward term based on the user's walking velocity.

However, a small format bi-treadmill as the Infinadeck cannot afford to have leeway to slowly change motor speed, as this would result in the user leaving the surface of the device. Jonghyun Kim et al. [3] brought key improvements to this approach by proposing two control modes. The standard mode works similarly to the one from [2], and the attenuated mode limits the treadmill accelerations to linear increases or decreases when the user is about to accelerate. This makes the algorithm suitable for much smaller treadmills without inducing excessive and destabilizing movement of the platform to the user. The main requirement is to keep the user within boundaries while imposing less intense accelerations.

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3 MOTION CONTROL IMPLEMENTATION

After calibrating the treadmill coordinate system, the vertical line on which the user center of gravity is estimated is the mean of the HMD and the backtracker position as illustrated in Fig. 1. This vertically projected position is referred to as $\vec{x}$, the users' position.

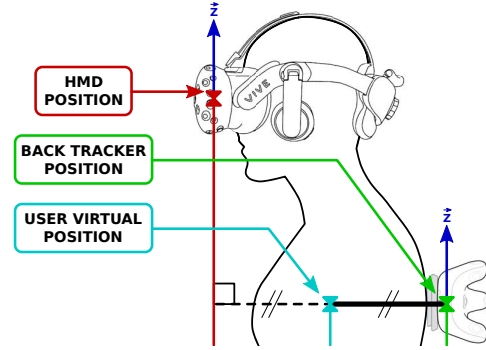


Figure 1: Schematic representation of the computation of the player position

Then, the implemented solution is sourced from [2]'s algorithm, but it is here used in 2D. With $\vec{u}_t$ the treadmill's speed at time t and $\vec{V}_w$ the user's walking speed, the floor projected kinematic equation is:

$$\dot{\vec{x}} = -\vec{u}_t + \vec{V}_w \quad (1)$$

With $(k_i)_{i \in \{1,3\}} \in \mathbb{R}^3$ the gains for the treadmill reaction to the user's input, the equation that aims to minimize the distance between the treadmill's surface center $\vec{x}_{\text{ref}}$ and the user position while aiming to reduce the anomalous force with the $\ddot{\vec{x}}$ factor [3] is:

$$\vec{0} = k_1 \cdot \ddot{\vec{x}} + k_2 \cdot \dot{\vec{x}} + k_3 \cdot (\vec{x} - \vec{x}_{\text{ref}}) \quad (2)$$

The combination of those equations (1 and 2) leads to the following formula for the treadmill speed $\vec{u}_t$:

$$\vec{u}_t = k_1 \cdot \ddot{\vec{x}} + (k_2 - 1) \cdot \dot{\vec{x}} + k_3 \cdot (\vec{x} - \vec{x}_{\text{ref}}) + \vec{V}_w \quad (3)$$

Still based on [2], the walking speed $\vec{V}_w$ was estimated using the same dynamic observer they used, as this value cannot be directly inferred from the raw inputs of the user's center position. Finally, a dead zone was added in the center of the treadmill to ensure that if the participant only slightly moved, the treadmill won't activate. In the rest of the paper, the anomalous force reduction approach is noted **AFR**.

4 EVALUATION METHODOLOGY

We designed an experimental protocol (Fig. 2) asking participants to perform 3 types of trajectories (straight line; trajectories including a single curved segment with a radius of either 0.5m, 1m, or 2m; turn in-place) with both methods in a pseudo-randomized order. After each individual trajectory trial, first, the perceived usability was

assessed by answering the question: "How easy was it to reach the destination?" between "Hard" and "Super easy" on a discrete scale between 0 and 10 with a step of 1. Second, the naturalness (i.e., the similarity of the walking experience with the regular walk) was assessed by answering the question: "How close was the walking experience to regular walking on the ground?", between "Different" and "The same". The EPFL ethical committee approved the protocol.

4.1 Setup

The Infinadeck omnidirectional treadmill [1] was provided with the software version: 2.3.3.3 and the firmware revision 3.0.5. The head-mounted display was connected to the computer's wireless module to allow infinite rotation. The provided ring used as a guardrail was removed, and instead, participants were safely harnessed with a professional harness attached to the ceiling, with a swivel to allow infinite rotations.

The variation of motion sickness throughout the experiment was assessed by asking each participant to fill out the Simulator Sickness Questionnaire (SSQ) [4] before and after the session. We formulated the following hypothesis :

1. H1 - The alternative algorithm (AFR) would outperform the original algorithm (O) on both usability and naturalness
2. H2 - Participant would not reach a severe level of motion sickness by the end of the experiment

4.2 Experimental protocol

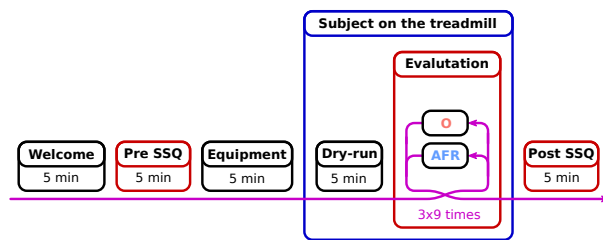


Figure 2: Schema of the protocol

Before the start of each condition, the subject was in a fully black environment while the treadmill placed them on the platform center. After the treadmill stabilized, the back screen faded out and revealed the path to the participant as illustrated in Fig. 3. Trajectories were always oriented to face the participant at the beginning of a condition. The void on the side implicitly forced subjects to follow the defined trajectories. A pilot study showed that the canopy played its role in preventing fear of heights.



Figure 3: Example of starting view for the participant when a curved trajectory is presented

5 ANALYSIS

The study involved the participation of 22 healthy participants, 26 years old on average (std: 8), of which 12 women (one could not do the experiment). No distinction was made between the curved trajectories with a non-null radius. Although the scores were filled on a discrete scale, the set of possible values was higher than 7 (11 values were proposed). Therefore we could reasonably consider the data as drawn from a continuous scale. The Shapiro test showed that all condition subsets were normally distributed for both the usability and naturalness scores hence allowing for parametric tests to be used (t-Test). The same analysis pipeline was used for each measured value (usability and naturalness). First, the two algorithms were compared for each type of measure using a two-sample two-sided t-test. Second, the effect size was measured using Cohen's d, and for significant differences, we performed two-sample one-sided t-test tests to assess the direction of the effect. Scores are displayed in Fig. 4.

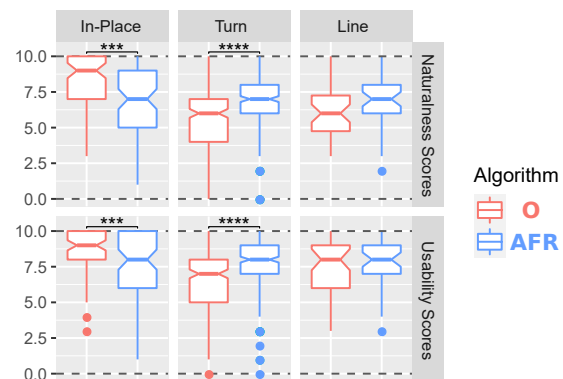


Figure 4: Scores for the usability and the naturalness per trajectory type and algorithm

6 DISCUSSION

Results show that AFR significantly performed better in terms of usability and naturalness for curved trajectories. No difference was observed on straight lines. On the other hand, O obtains significantly better scores for the naturalness of in-place trajectories. Indeed, when one moves for the in-place condition, one expects the treadmill to stay still. In the original algorithm, if the player moves slightly outside of the dead zone, the reaction is slight, whereas, with our implementation of the AFR, the control induces a small discontinuity. Finally, it was observed that walking on the treadmill did not fully alleviate the motion sickness as the post-SSQ score highlighted a significantly different level of motion sickness compared to the Pre SSQ scores. However, the final scores corresponded to the appearance of only minor symptoms.

REFERENCES

- [1] Infinadeck, "Infinadeck." <https://www.infinadeck.com/>, 2018.
- [2] J. L. Souman, P. R. Giordano, I. Frissen, A. de Luca, and M. O. Ernst, "Making virtual walking real: Perceptual evaluation of a new treadmill control algorithm," 2010.
- [3] J. Kim, H.-S. Park, and D. L. Damiano, "An interactive treadmill under a novel control scheme for simulating overground walking by reducing anomalous force," *IEEE/ASME Transactions on Mechatronics*, vol. 20, no. 3, pp. 1491–1496, 2015.
- [4] R. S. Kennedy, N. E. Lane, K. S. Berbaum, and M. G. Lilienthal, "Simulator sickness questionnaire: An enhanced method for quantifying simulator sickness," *The international journal of aviation psychology*, vol. 3, no. 3, pp. 203–220, 1993.